Consider the following database schema for an E-commerce system:

```
mysql> Create database 054_E_Commerce;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| 054_e_commerce     |
| esports_db         |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.00 sec)
```

```
mysql> use 054_E_Commerce;
Database changed
```

Tables:

Customers (customer_id, name, email, city)

Orders (order_id, customer_id, order_date, total_amount)

Products (product_id, product_name, category, price)

Order_Items (order_item_id, order_id, product_id, quantity, subtotal)

Sellers (seller_id, seller_name, city)

```
mysql> CREATE TABLE Customers_054(
    ->     customer_id INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     email VARCHAR(100),
    ->     city VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE Orders_054(
    ->     order_id INT PRIMARY KEY,
    ->     customer_id INT,
    ->     order_date DATE,
    ->     total_amount DECIMAL(10,2),
    ->     FOREIGN KEY (customer_id) REFERENCES Customers_054(customer_id)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE Products_054 (
    ->     product_id INT PRIMARY KEY,
    ->     product_name VARCHAR(50),
    ->     category VARCHAR(50),
    ->     price DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE Order_Items_054 (
    ->     order_item_id INT PRIMARY KEY,
    ->     order_id INT,
    ->     product_id INT,
    ->     quantity INT,
    ->     subtotal DECIMAL(10,2),
    ->     FOREIGN KEY (order_id) REFERENCES Orders_054(order_id),
    ->     FOREIGN KEY (product_id) REFERENCES Products_054(product_id)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE Sellers_054(
    ->     seller_id INT PRIMARY KEY,
    ->     seller_name VARCHAR(50),
    ->     city VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> show tables;
+-------------------------+
| Tables_in_054_e_commerce |
+-------------------------+
| customers_054           |
| order_items_054         |
| orders_054              |
| products_054            |
| sellers_054             |
+-------------------------+
5 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Customers_054 VALUES
    -> (1, 'Amit Sharma', 'amit@gmail.com', 'Pune'),
    -> (2, 'Priya Patel', 'priya@gmail.com', 'Mumbai'),
    -> (3, 'Rahul Verma', 'rahul@gmail.com', 'Delhi'),
    -> (4, 'Sneha Iyer', 'sneha@gmail.com', 'Chennai'),
    -> (5, 'Arjun Reddy', 'arjun@gmail.com', 'Hyderabad'),
    -> (6, 'Neha Gupta', 'neha@gmail.com', 'Pune');
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Orders_054 VALUES
    -> (101, 1, '2026-03-01', 50000),
    -> (102, 2, '2026-02-15', 2000),
    -> (103, 1, '2026-01-10', 1500),
    -> (104, 3, '2026-03-10', 7000),
    -> (105, 3, '2025-12-25', 12000),
    -> (106, 5, '2026-03-18', 3000);
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Products_054 VALUES
    -> (1, 'Laptop', 'Electronics', 50000),
    -> (2, 'Mobile', 'Electronics', 20000),
    -> (3, 'Headphones', 'Accessories', 1500),
    -> (4, 'Keyboard', 'Accessories', 800),
    -> (5, 'Shoes', 'Fashion', 2500),
    -> (6, 'Watch', 'Fashion', 3000),
    -> (7, 'Tablet', 'Electronics', 15000);
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Order_Items_054 VALUES
    -> (1, 101, 1, 1, 50000),
    -> (2, 102, 3, 2, 3000),
    -> (3, 103, 4, 1, 800),
    -> (4, 104, 2, 1, 20000),
    -> (5, 104, 3, 2, 3000),
    -> (6, 105, 5, 2, 5000),
    -> (7, 106, 6, 1, 3000);
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Sellers_054 VALUES
    -> (1, 'Raj Electronics', 'Mumbai'),
    -> (2, 'Tech World', 'Delhi'),
    -> (3, 'Fashion Hub', 'Pune'),
    -> (4, 'Gadget Store', 'Bangalore'),
    -> (5, 'Style Zone', 'Hyderabad');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> Select * from Customers_054;
+-------------+-------------+-----------------+-----------+
| customer_id | name        | email           | city      |
+-------------+-------------+-----------------+-----------+
|           1 | Amit Sharma | amit@gmail.com  | Pune      |
|           2 | Priya Patel | priya@gmail.com | Mumbai    |
|           3 | Rahul Verma | rahul@gmail.com | Delhi     |
|           4 | Sneha Iyer  | sneha@gmail.com | Chennai   |
|           5 | Arjun Reddy | arjun@gmail.com | Hyderabad |
|           6 | Neha Gupta  | neha@gmail.com  | Pune      |
+-------------+-------------+-----------------+-----------+
6 rows in set (0.00 sec)
```

```
mysql> Select * from Orders_054;
+----------+-------------+------------+--------------+
| order_id | customer_id | order_date | total_amount |
+----------+-------------+------------+--------------+
|      101 |           1 | 2026-03-01 |     50000.00 |
|      102 |           2 | 2026-02-15 |      2000.00 |
|      103 |           1 | 2026-01-10 |      1500.00 |
|      104 |           3 | 2026-03-10 |      7000.00 |
|      105 |           3 | 2025-12-25 |     12000.00 |
|      106 |           5 | 2026-03-18 |      3000.00 |
+----------+-------------+------------+--------------+
6 rows in set (0.00 sec)
```

```
mysql> Select * from Products_054;
+------------+--------------+-------------+----------+
| product_id | product_name | category    | price    |
+------------+--------------+-------------+----------+
|          1 | Laptop       | Electronics | 50000.00 |
|          2 | Mobile       | Electronics | 20000.00 |
|          3 | Headphones   | Accessories |  1500.00 |
|          4 | Keyboard     | Accessories |   800.00 |
|          5 | Shoes        | Fashion     |  2500.00 |
|          6 | Watch        | Fashion     |  3000.00 |
|          7 | Tablet       | Electronics | 15000.00 |
+------------+--------------+-------------+----------+
7 rows in set (0.00 sec)
```

```
mysql> Select * from Order_Items_054;
+---------------+----------+------------+----------+----------+
| order_item_id | order_id | product_id | quantity | subtotal |
+---------------+----------+------------+----------+----------+
|             1 |      101 |          1 |        1 | 50000.00 |
|             2 |      102 |          3 |        2 |  3000.00 |
|             3 |      103 |          4 |        1 |   800.00 |
|             4 |      104 |          2 |        1 | 20000.00 |
|             5 |      104 |          3 |        2 |  3000.00 |
|             6 |      105 |          5 |        2 |  5000.00 |
|             7 |      106 |          6 |        1 |  3000.00 |
+---------------+----------+------------+----------+----------+
7 rows in set (0.00 sec)
```

```
mysql> Select * from Sellers_054;
+-----------+-----------------+-----------+
| seller_id | seller_name     | city      |
+-----------+-----------------+-----------+
|         1 | Raj Electronics | Mumbai    |
|         2 | Tech World      | Delhi     |
|         3 | Fashion Hub     | Pune      |
|         4 | Gadget Store    | Bangalore |
|         5 | Style Zone      | Hyderabad |
+-----------+-----------------+-----------+
5 rows in set (0.00 sec)
```

QUERIES :

1. Retrieve all customers along with their corresponding orders (including customers who haven't ordered).

```
SELECT c.*, o.*
   -> FROM  Customers_054 c
   -> LEFT JOIN Orders_054 o
   -> ON c.customer_id = o.customer_id;
```

```
mysql> SELECT c.*, o.*
    -> FROM  Customers_054 c
    -> LEFT JOIN Orders_054 o
    -> ON c.customer_id = o.customer_id;
+-------------+-------------+-----------------+-----------+----------+-------------+------------+--------------+
| customer_id | name        | email           | city      | order_id | customer_id | order_date | total_amount |
+-------------+-------------+-----------------+-----------+----------+-------------+------------+--------------+
|           1 | Amit Sharma | amit@gmail.com  | Pune      |      101 |           1 | 2026-03-01 |     50000.00 |
|           1 | Amit Sharma | amit@gmail.com  | Pune      |      103 |           1 | 2026-01-10 |      1500.00 |
|           2 | Priya Patel | priya@gmail.com | Mumbai    |      102 |           2 | 2026-02-15 |      2000.00 |
|           3 | Rahul Verma | rahul@gmail.com | Delhi     |      104 |           3 | 2026-03-10 |      7000.00 |
|           3 | Rahul Verma | rahul@gmail.com | Delhi     |      105 |           3 | 2025-12-25 |     12000.00 |
|           4 | Sneha Iyer  | sneha@gmail.com | Chennai   |     NULL |        NULL | NULL       |         NULL |
|           5 | Arjun Reddy | arjun@gmail.com | Hyderabad |      106 |           5 | 2026-03-18 |      3000.00 |
|           6 | Neha Gupta  | neha@gmail.com  | Pune      |     NULL |        NULL | NULL       |         NULL |
+-------------+-------------+-----------------+-----------+----------+-------------+------------+--------------+
8 rows in set (0.00 sec)
```

2. List all orders along with the product names and their quantities.

SELECT o.order_id, p.product_name, oi.quantity

-> FROM Orders_054 o

-> JOIN Order_Items_054 oi ON o.order_id = oi.order_id

-> JOIN Products_054 p ON oi.product_id = p.product_id;

```
mysql> SELECT o.order_id, p.product_name, oi.quantity
    -> FROM Orders_054 o
    -> JOIN Order_Items_054 oi ON o.order_id = oi.order_id
    -> JOIN Products_054 p ON oi.product_id = p.product_id;
+----------+--------------+----------+
| order_id | product_name | quantity |
+----------+--------------+----------+
|      101 | Laptop       |        1 |
|      103 | Keyboard     |        1 |
|      102 | Headphones   |        2 |
|      104 | Mobile       |        1 |
|      104 | Headphones   |        2 |
|      105 | Shoes        |        2 |
|      106 | Watch        |        1 |
+----------+--------------+----------+
7 rows in set (0.00 sec)
```

3. Find the total number of orders placed by each customer.

SELECT c.name, COUNT(o.order_id) AS total_orders_054

-> FROM Customers_054 c

-> LEFT JOIN Orders_054 o ON c.customer_id = o.customer_id

-> GROUP BY c.name;

```
mysql> SELECT c.name, COUNT(o.order_id) AS total_orders_054
    -> FROM Customers_054 c
    -> LEFT JOIN Orders_054 o ON c.customer_id = o.customer_id
    -> GROUP BY c.name;
+-------------+------------------+
| name        | total_orders_054 |
+-------------+------------------+
| Amit Sharma |                2 |
| Priya Patel |                1 |
| Rahul Verma |                2 |
| Sneha Iyer  |                0 |
| Arjun Reddy |                1 |
| Neha Gupta  |                0 |
+-------------+------------------+
6 rows in set (0.00 sec)
```

4. Find the total number of products available in each category.

SELECT category, COUNT(*) AS total_products_054

-> FROM Products_054

-> GROUP BY category;

```
mysql> SELECT category, COUNT(*) AS total_products_054
    -> FROM Products_054
    -> GROUP BY category;
+-------------+--------------------+
| category    | total_products_054 |
+-------------+--------------------+
| Electronics |                  3 |
| Accessories |                  2 |
| Fashion     |                  2 |
+-------------+--------------------+
3 rows in set (0.00 sec)
```

5. Retrieve the order details, including customer name and total amount, for all orders placed in the last 30 days.

SELECT o.order_id, c.name, o.total_amount

-> FROM Orders_054 o

-> JOIN Customers_054 c ON o.customer_id = c.customer_id

-> WHERE o.order_date >= CURRENT_DATE - INTERVAL 30 DAY;

```
mysql> SELECT o.order_id, c.name, o.total_amount
    -> FROM Orders_054 o
    -> JOIN Customers_054 c ON o.customer_id = c.customer_id
    -> WHERE o.order_date >= CURRENT_DATE - INTERVAL 30 DAY;
+----------+-------------+--------------+
| order_id | name        | total_amount |
+----------+-------------+--------------+
|      101 | Amit Sharma |     50000.00 |
|      104 | Rahul Verma |      7000.00 |
|      106 | Arjun Reddy |      3000.00 |
+----------+-------------+--------------+
3 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Product_Sellers_054 (
    ->     seller_id INT,
    ->     product_id INT,
    ->     PRIMARY KEY (seller_id, product_id),
    ->     FOREIGN KEY (seller_id) REFERENCES Sellers_054(seller_id),
    ->     FOREIGN KEY (product_id) REFERENCES Products_054(product_id)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> show tables;
+-------------------------+
| Tables_in_054_e_commerce |
+-------------------------+
| customers_054           |
| order_items_054         |
| orders_054              |
| product_sellers_054     |
| products_054            |
| sellers_054             |
+-------------------------+
6 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Product_Sellers_054 VALUES
    -> (1, 1),
    -> (2, 1),
    -> (1, 2),
    -> (4, 2),
    -> (2, 3),
    -> (3, 5),
    -> (5, 6),
    -> (4, 7);
Query OK, 8 rows affected (0.01 sec)
```

6. Retrieve the names of sellers who sell a specific product (e.g., "Laptop").

SELECT s.seller_name

-> FROM Sellers_054 s

-> JOIN Product_Sellers_054 ps ON s.seller_id = ps.seller_id

-> JOIN Products_054 p ON ps.product_id = p.product_id

-> WHERE p.product_name = 'Laptop';

```
mysql> SELECT s.seller_name
    -> FROM Sellers_054 s
    -> JOIN Product_Sellers_054 ps ON s.seller_id = ps.seller_id
    -> JOIN Products_054 p ON ps.product_id = p.product_id
    -> WHERE p.product_name = 'Laptop';
+-----------------+
| seller_name     |
+-----------------+
| Raj Electronics |
| Tech World      |
+-----------------+
2 rows in set (0.01 sec)
```

7. Show all customers who have never placed an order.

SELECT c.name

-> FROM Customers_054 c

-> LEFT JOIN Orders_054 o ON c.customer_id = o.customer_id

-> WHERE o.order_id IS NULL;

```
mysql> SELECT c.name
    -> FROM Customers_054 c
    -> LEFT JOIN Orders_054 o ON c.customer_id = o.customer_id
    -> WHERE o.order_id IS NULL;
+------------+
| name       |
+------------+
| Sneha Iyer |
| Neha Gupta |
+------------+
2 rows in set (0.00 sec)
```

8. Retrieve details of orders where the total amount is greater than the average order total.(solve using subquery)

SELECT *

-> FROM Orders_054

-> WHERE total_amount > (

-> SELECT AVG(total_amount) FROM Orders_054

->);

```
mysql> SELECT *
    -> FROM Orders_054
    -> WHERE total_amount > (
    ->     SELECT AVG(total_amount) FROM Orders_054
    -> );
+----------+-------------+------------+--------------+
| order_id | customer_id | order_date | total_amount |
+----------+-------------+------------+--------------+
|      101 |           1 | 2026-03-01 |     50000.00 |
+----------+-------------+------------+--------------+
1 row in set (0.00 sec)
```

9. Find customers who have placed at least two orders.

SELECT customer_id

-> FROM Orders_054

-> GROUP BY customer_id

-> HAVING COUNT(order_id) >= 2;

```
mysql> SELECT customer_id
    -> FROM Orders_054
    -> GROUP BY customer_id
    -> HAVING COUNT(order_id) >= 2;
+-------------+
| customer_id |
+-------------+
|           1 |
|           3 |
+-------------+
2 rows in set (0.00 sec)
```

10. Find the top 3 most ordered products based on quantity sold.

SELECT p.product_name, SUM(oi.quantity) AS total_quantity

-> FROM Order_Items_054 oi

-> JOIN Products_054 p ON oi.product_id = p.product_id

-> GROUP BY p.product_name

-> ORDER BY total_quantity DESC

-> LIMIT 3;

```
mysql> SELECT p.product_name, SUM(oi.quantity) AS total_quantity
    -> FROM Order_Items_054 oi
    -> JOIN Products_054 p ON oi.product_id = p.product_id
    -> GROUP BY p.product_name
    -> ORDER BY total_quantity DESC
    -> LIMIT 3;
+--------------+----------------+
| product_name | total_quantity |
+--------------+----------------+
| Headphones   |              4 |
| Shoes        |              2 |
| Laptop       |              1 |
+--------------+----------------+
3 rows in set (0.00 sec)
```

11. Display the product names that are sold by multiple sellers.

SELECT p.product_name

-> FROM Product_Sellers_054 ps

-> JOIN Products_054 p ON ps.product_id = p.product_id

-> GROUP BY p.product_name

-> HAVING COUNT(DISTINCT ps.seller_id) > 1;

```
mysql> SELECT p.product_name
    -> FROM Product_Sellers_054 ps
    -> JOIN Products_054 p ON ps.product_id = p.product_id
    -> GROUP BY p.product_name
    -> HAVING COUNT(DISTINCT ps.seller_id) > 1;
+--------------+
| product_name |
+--------------+
| Laptop       |
| Mobile       |
+--------------+
2 rows in set (0.00 sec)
```

12. Retrieve sellers who have never sold any product.

SELECT s.seller_name

-> FROM Sellers_054 s

-> LEFT JOIN Product_Sellers_054 ps ON s.seller_id = ps.seller_id

-> WHERE ps.product_id IS NULL;

```
mysql> SELECT s.seller_name
    -> FROM Sellers_054 s
    -> LEFT JOIN Product_Sellers_054 ps ON s.seller_id = ps.seller_id
    -> WHERE ps.product_id IS NULL;
Empty set (0.00 sec)
```

13. Find all products that have never been ordered.

SELECT p.product_name

-> FROM Products_054 p

-> LEFT JOIN Order_Items_054 oi ON p.product_id = oi.product_id

-> WHERE oi.product_id IS NULL;

```
mysql> SELECT p.product_name
    -> FROM Products_054 p
    -> LEFT JOIN Order_Items_054 oi ON p.product_id = oi.product_id
    -> WHERE oi.product_id IS NULL;
+--------------+
| product_name |
+--------------+
| Tablet       |
+--------------+
1 row in set (0.00 sec)
```

14. Retrieve the names of customers who have placed the highest number of orders.

SELECT name

-> FROM Customers_054

-> WHERE customer_id IN (

-> SELECT customer_id

-> FROM Orders_054

-> GROUP BY customer_id

-> HAVING COUNT(order_id) = (

-> SELECT MAX(order_count)

-> FROM (

-> SELECT COUNT(*) AS order_count

-> FROM Orders_054

-> GROUP BY customer_id

->) t

->)

->);

```
mysql> SELECT name
    -> FROM Customers_054
    -> WHERE customer_id IN (
    ->     SELECT customer_id
    ->     FROM Orders_054
    ->     GROUP BY customer_id
    ->     HAVING COUNT(order_id) = (
    ->         SELECT MAX(order_count)
    ->         FROM (
    ->             SELECT COUNT(*) AS order_count
    ->             FROM Orders_054
    ->             GROUP BY customer_id
    ->         ) t
    ->     )
    -> );
+-------------+
| name        |
+-------------+
| Amit Sharma |
| Rahul Verma |
+-------------+
2 rows in set (0.00 sec)
```

15. Find all customers who have ordered more than 5 different products.

(Hint: Use COUNT DISTINCT on product_id.)

SELECT c.name

-> FROM Customers_054 c

-> JOIN Orders_054 o ON c.customer_id = o.customer_id

-> JOIN Order_Items_054 oi ON o.order_id = oi.order_id

-> GROUP BY c.name

-> HAVING COUNT(DISTINCT oi.product_id) > 5;

```
mysql> SELECT c.name
    -> FROM Customers_054 c
    -> JOIN Orders_054 o ON c.customer_id = o.customer_id
    -> JOIN Order_Items_054 oi ON o.order_id = oi.order_id
    -> GROUP BY c.name
    -> HAVING COUNT(DISTINCT oi.product_id) > 5;
Empty set (0.00 sec)
```

16. Find products that are sold by at least two different sellers but have never been ordered. (Hint: Use HAVING COUNT(DISTINCT seller_id) > 1 and NOT EXISTS.)

SELECT p.product_name

-> FROM Products_054 p

```
-> WHERE p.product_id IN (
->     SELECT ps.product_id
->     FROM Product_Sellers_054 ps
->     GROUP BY ps.product_id
->     HAVING COUNT(DISTINCT ps.seller_id) > 1
-> )
-> AND p.product_id NOT IN (
->     SELECT product_id FROM Order_Items_054
-> );
```

```
mysql> SELECT p.product_name
    -> FROM Products_054 p
    -> WHERE p.product_id IN (
    ->     SELECT ps.product_id
    ->     FROM Product_Sellers_054 ps
    ->     GROUP BY ps.product_id
    ->     HAVING COUNT(DISTINCT ps.seller_id) > 1
    -> )
    -> AND p.product_id NOT IN (
    ->     SELECT product_id FROM Order_Items_054
    -> );
Empty set (0.00 sec)
```

17. Find the customer who has spent the most money overall.

(Hint: Use SUM and ORDER BY.)

SELECT c.name, SUM(o.total_amount) AS total_spent

-> FROM Customers_054 c

-> JOIN Orders_054 o ON c.customer_id = o.customer_id

-> GROUP BY c.name

-> ORDER BY total_spent DESC

-> LIMIT 1;

```
mysql> SELECT c.name, SUM(o.total_amount) AS total_spent
    -> FROM Customers_054 c
    -> JOIN Orders_054 o ON c.customer_id = o.customer_id
    -> GROUP BY c.name
    -> ORDER BY total_spent DESC
    -> LIMIT 1;
+-------------+-------------+
| name        | total_spent |
+-------------+-------------+
| Amit Sharma |    51500.00 |
+-------------+-------------+
1 row in set (0.00 sec)
```

18. Find all customers who have either placed an order or live in the same city as a seller. (Hint: Use UNION to combine customers who placed orders and those in seller cities.)

SELECT name FROM Customers_054

-> WHERE customer_id IN (SELECT customer_id FROM Orders_054)

->

-> UNION

->

-> SELECT name FROM Customers_054

-> WHERE city IN (SELECT city FROM Sellers_054);

```
mysql> SELECT name FROM Customers_054
    -> WHERE customer_id IN (SELECT customer_id FROM Orders_054)
    ->
    -> UNION
    ->
    -> SELECT name FROM Customers_054
    -> WHERE city IN (SELECT city FROM Sellers_054);
+-------------+
| name        |
+-------------+
| Amit Sharma |
| Priya Patel |
| Rahul Verma |
| Arjun Reddy |
| Neha Gupta  |
+-------------+
5 rows in set (0.00 sec)
```

19. Retrieve all products that are either in stock with at least one seller or have been ordered at least once.(Hint: Use UNION between Product_Sellers and Order_Items.)

SELECT product_id FROM Product_Sellers_054

-> UNION

-> SELECT product_id FROM Order_Items_054;

```
mysql> SELECT product_id FROM Product_Sellers_054
    ->
    -> UNION
    ->
    -> SELECT product_id FROM Order_Items_054;
+------------+
| product_id |
+------------+
|          1 |
|          2 |
|          3 |
|          5 |
|          6 |
|          7 |
|          4 |
+------------+
7 rows in set (0.00 sec)
```

20. Retrieve products that have been both ordered and are currently in stock.

(Hint: Use INTERSECT between Order_Items and Product_Sellers.)

SELECT product_id FROM Product_Sellers_054

-> INTERSECT

-> SELECT product_id FROM Order_Items_054;

```
mysql> SELECT product_id FROM Product_Sellers_054
    ->
    -> INTERSECT
    ->
    -> SELECT product_id FROM Order_Items_054;
+------------+
| product_id |
+------------+
|          1 |
|          2 |
|          3 |
|          5 |
|          6 |
+------------+
5 rows in set (0.00 sec)
```

21. Find customers who have both placed an order and live in a city where a seller is located. (Hint: Use INTERSECT between customers who have placed orders and customers who live in a seller's city.)

SELECT customer_id FROM Orders_054

-> INTERSECT

-> SELECT customer_id FROM Customers_054

-> WHERE city IN (SELECT city FROM Sellers_054);

```
mysql> SELECT customer_id FROM Orders_054
    ->
    -> INTERSECT
    ->
    -> SELECT customer_id FROM Customers_054
    -> WHERE city IN (SELECT city FROM Sellers_054);
+-------------+
| customer_id |
+-------------+
|           1 |
|           2 |
|           3 |
|           5 |
+-------------+
4 rows in set (0.00 sec)
```

22. Retrieve all customers who have placed at least one order in each year available in the Database. (Hint: Use INTERSECT with orders grouped by year.)

SELECT customer_id

-> FROM Orders_054

-> GROUP BY customer_id

```
-> HAVING COUNT(DISTINCT YEAR(order_date)) = (
->     SELECT COUNT(DISTINCT YEAR(order_date))
->     FROM Orders_054
-> );
```

```
mysql> SELECT customer_id
    -> FROM Orders_054
    -> GROUP BY customer_id
    -> HAVING COUNT(DISTINCT YEAR(order_date)) = (
    ->     SELECT COUNT(DISTINCT YEAR(order_date))
    ->     FROM Orders_054
    -> );
+-------------+
| customer_id |
+-------------+
|           3 |
+-------------+
1 row in set (0.00 sec)
```